Review_App_Snapshot.pdf

GenAI Case Study: Customer Sentiment & Review Response Automation
Milestone 3 — GenAI-Powered Review Assistant (Deployment Phase)

Business Question: "How can we respond faster while staying safe and consistent?"

Tool: Gradio app powered by a packaged NLP/sentiment pipeline + Gemini
Prepared by: Dev

Executive Summary

This milestone operationalizes the insights from Milestones 1 and 2 into a working assistant. The trained NLP pipeline (text cleaning, sentiment scoring, urgency scoring, theme extraction) was packaged into a single reusable class, saved with pickle, and wired into a Gradio web app that also calls Gemini to draft a policy-aligned response for agent review. The tool assists customer support agents rather than replacing them — every draft response is explicitly labeled for human review before sending (Human-in-the-Loop).

Component	Status
Pipeline packaged (pickle)	Complete — sentiment_pipeline.pkl
Gradio app built	Complete — 4 outputs: sentiment, urgency, themes, draft response
Database integration (optional)	Complete — SQLite round-trip, generated_responses table
App snapshots	3 examples covering Negative, Neutral, and Positive segments

1. Packaging the Trained NLP and ML Pipeline

All logic from Milestones 1 and 2 — text preprocessing, the VADER-based sentiment model, urgency scoring, and a new keyword-based theme extractor — was bundled into one Python class, **ReviewAnalyzer**, with a single **analyze(title, text)** method returning sentiment, sentiment score, urgency score, and detected themes. The class instance was saved with pickle (**sentiment_pipeline.pkl**) and successfully reloaded for inference, confirming the pipeline is portable and ready for deployment.

Test Result (Pipeline Verification)

Review: “Not as described” — “This Webcam HD is defective and unusable. Cheapest materials, worst functionality. I want a full refund! Very angry customer!”

```
{'sentiment': 'Negative', 'sentiment_score': -0.897, 'urgency_score': 4, 'themes': ['Product Defect', 'Refund Request', 'Quality Complaint']}
```

The reloaded pipeline produced an identical result to the original in-session pipeline, confirming successful save/load.

2. Gradio Application

A Gradio Blocks app was built accepting Review Title and Review Text as input, and returning four outputs: Predicted Sentiment, Urgency Score, Detected Themes, and a GenAI Draft Response. The app displays a persistent note reinforcing the Human-in-the-Loop principle: draft responses are for agent review, not automatic sending.

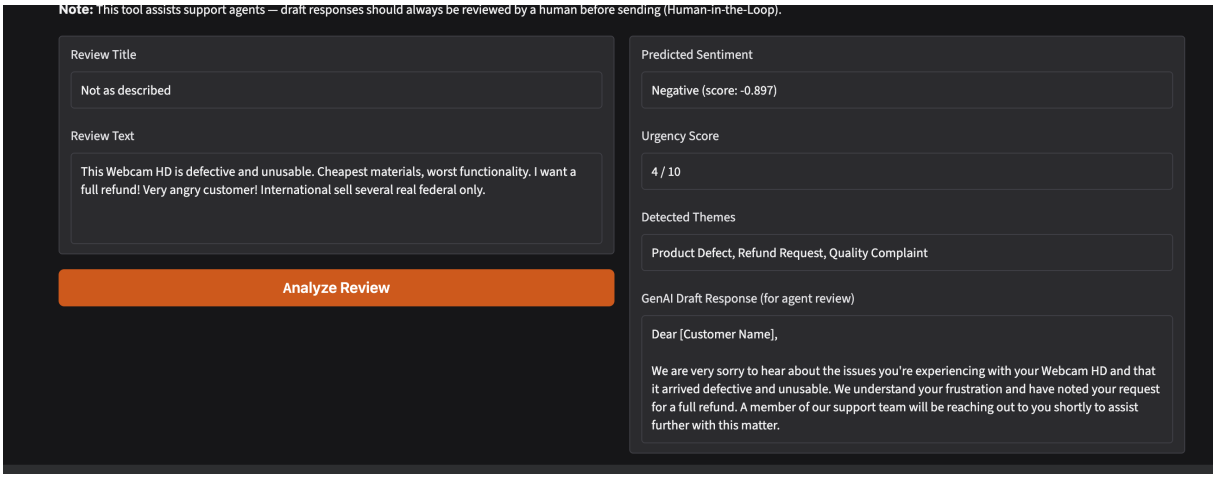


Figure 1: App output for a Negative review (Webcam HD, defective/refund request).

3. Database Integration (Optional)

As an optional extension, the app was connected back to the SQLite reviews database. A random review was pulled directly from **customer_reviews**, run through the same packaged pipeline and Gemini draft-response generator, and the result was written to a new **generated_responses** table — keeping the original review data untouched.

Field	Value
Review # (rowid)	968
Rating	4
Title	Absolutely fantastic!
Text	This Desk Lamp is exactly what I needed. Great quality, fast shipping, and the build quality works perfectly. Very satisfied customer here!
Sentiment	Positive
Urgency Score	0
Themes	Praise
Draft Response	Thank you for your fantastic review! We're delighted to hear that our Desk Lamp is exactly what you needed and that you're so satisfied with your purchase. Our team is always here to assist should you have any questions or require further assistance.

The round-trip was verified by querying the **generated_responses** table after insertion, confirming the row was written correctly and the original review data was left unmodified.

4. App Snapshots — 3 Real Examples

To demonstrate the assistant across the full sentiment spectrum, three real reviews were run through the tool: one Negative, one Neutral, and one Positive (via the database integration above).

Segment	Product	Sentiment	Urgency	Themes
Negative	Webcam HD	Negative (-0.897)	4	Product Defect, Refund Request, Quality Complaint
Neutral	Smart Watch	Positive (0.341)*	0	Refund Request*
Positive	Desk Lamp	Positive (via DB)	0	Praise

*See known limitation noted below.

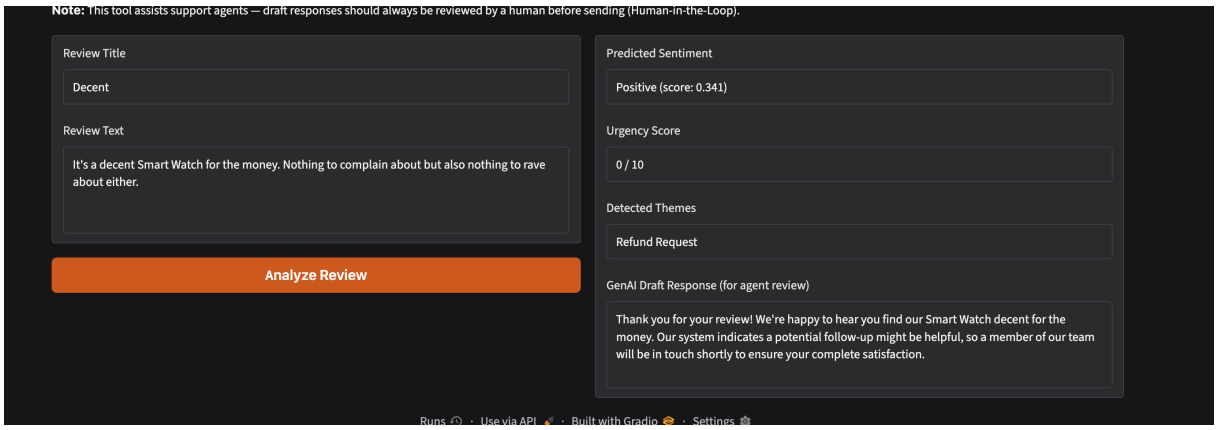


Figure 2: App output for a Neutral review (Smart Watch, “decent... for the money”).

Known Limitation: False-Positive Theme Detection

The Neutral (Smart Watch) example above surfaced a real limitation in the keyword-based theme detector: the review text “It's a decent Smart Watch for the money...” was incorrectly tagged with the “Refund Request” theme. This happened because the theme keyword list includes the word “money” (originally chosen because it appears in genuine refund complaints like “want my money back”), but here it appears in the harmless, common phrase “for the money” (i.e., value-for-price), with no actual refund request present.

Business implication: This is a concrete, real example of why the guidelines specify a Human-in-the-Loop design rather than auto-routing or auto-responding based on AI-detected themes. A support agent reviewing this flagged theme would immediately recognize it as a false positive and correct the routing, whereas a fully automated system might have incorrectly escalated a satisfied customer as a refund case. This is documented here rather than silently fixed, as a genuine illustration of the tool's current limitations and the value of the human review step.

5. Summary

- The trained pipeline (cleaning, sentiment, urgency, theme extraction) was successfully packaged, saved, and reloaded for inference, confirming production-readiness of the packaging step.
- The Gradio app successfully surfaces sentiment, urgency, themes, and a GenAI-drafted response for any review a support agent pastes in, with the Human-in-the-Loop principle visibly built into the UI.
- Optional database integration confirmed a working round-trip: pulling a live review, running predictions, and storing the generated response back without modifying the original data.
- Testing across Negative, Neutral, and Positive examples surfaced a real limitation (a generic keyword causing a false-positive theme) — documented rather than hidden, reinforcing why AI-generated output in this system is always framed as a draft for human review, not an automatic action.